

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 40%

Annexure A

Coding challenge

85%

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1. Develop a Python program that generates a Star Topology for a given number of hosts connected to a central switch. The program should accept the number of hosts as input and display the topology either in ASCII format or a graphical representation. For each host, print the host label, the name of the central switch, and the corresponding Cat6 UTP cable length connecting the host to the switch. Verify that all hosts are connected through the central switch and explain how the star topology facilitates reliable communication in a local area network.
2. Write a Python function named `validate_segment(length_m)` to verify whether a given **UTP cable segment** satisfies the IEEE 802.3 Ethernet standard, which specifies a maximum cable length of **100 meters**. The function should return **True** for valid cable lengths and return **False** along with an appropriate error message for invalid lengths exceeding the permissible limit. Demonstrate the correctness of the program by executing it with at least **three different test cases** and interpret the results.

RUBRICS FOR CALCULATION

	Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A	Problem Understanding	15% 1.5	Fully understands the problem and requirements; no clarification needed	Understands most requirements with minor clarification	Partial understanding; misses some requirements	Poor understanding of the problem
B	Algorithm Design	20% 2	Efficient, optimal, and well-structured algorithm	Correct algorithm with minor inefficiencies	Basic approach but not optimal	Incorrect or no clear algorithm
C	Code Quality	15% 1.5	Clean, well-organized, readable, and properly formatted code	Mostly clean code with minor issues	Code is somewhat unclear or inconsistent	Poorly written, unreadable code
D	Functionality	20% 2	Code works perfectly for all test cases	Works for most test cases	Works for limited cases	Does not run or fails major cases
E	Error Handling	10% 1	Handles all edge cases and errors effectively	Handles some edge cases	Limited error handling	No error handling
F	Efficiency	10% 1	Highly optimized in time and space complexity	Reasonably efficient	Some inefficiencies	Highly inefficient
G	Documentation & Comments	5% 0.5	Clear and meaningful comments and documentation	Adequate comments	Few or unclear comments	No comments
H	Testing & Debugging	5% 0.5	Thoroughly tested with multiple cases; no bugs	Tested with some cases	Minimal testing	No testing done

	A	B	C	D	E	F	G	H	
Q1	1	2	1.5	2	1	1	0.5	0.5	9.5
Q2	1	2	1.5	2	1	1	0.5	0.5	9.5
Q3	1.5	2	1	1.5	0.5	0.5	0.5	0.5	8
Q4	1.5	1.5	1	1.5	0.5	1	0.5	0.5	8
Q5	1	1	1	2	1	0.5	0.5	0.5	7.5

A2.5